

Plant passengers

NOT ALL PLANTS that live on others are parasites (pp. 44–45). In fact, many more of them are simply passengers that grow on larger plants, such as trees, without causing them any harm. Such plants are described as epiphytic, and many of them can get all the water they need simply by absorbing it from the air, or by collecting it in structures formed for the purpose. They collect minerals by extracting them from trickling rainwater and plant debris. Being an epiphyte gives a small plant a chance to collect a lot of light without the need for tall stems. So successful is this way of life that few trees are without their passengers. In cool parts of the world, epiphytes are usually small, simple plants such as algae, lichens, and mosses. But in moist regions close to the equator, they are much larger. In addition to the plants that spend their entire lives up in the trees, there are others that start or end their lives in this

way. Some creeping plants, known as stranglers, germinate on trees and then become rooted in the soil. Others climb up on to plants but then their roots wither away, leaving them perched high up near the light.



FOREST CLIMBERS
These large, woody climbers, known as lianas, grow in the forests of Central America.

Leaves have a special coating to reduce water loss



High up in the trees it is easier for plants to collect enough light

HIGH-RISE FLOWERS
In the forests of Sri Lanka, epiphytic orchids can be found growing on the trees. Most epiphytic plants live in tropical and subtropical forests, because they require humid conditions.



THE BROMELIAD'S PRIVATE POND
Bromeliads are a family of plant that includes the pineapple. Many bromeliads grow on other plants. Instead of collecting water with long, aerial roots like orchids, they channel rainwater into a central reservoir (right) with their stiff, spiky leaves. Hairs on the leaves then absorb the water so the plant can use it. A big bromeliad holds well over 1 gallon (5 liters) of water and provides a home for water animals such as tree-frog tadpoles.

